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System and method to perform in-ear noise dosimetry and personal attenuation rating under an electro-acoustic earplug while excluding wearer-induced disturbances and separating exposure sources

Abstract

A system and method to measure noise reduction and evaluate the contributions of various sound sources to noise exposure dose of exposition using electro-acoustic earplugs is provided. The system may be implemented as an advanced HPD in the form of an electro-acoustical earplug. The earplug comprises an OEM **10** and an IEM. The system is configured to calculate an estimated IEM signal based on calculation of estimates of residual ambient noise, payback sounds or WID present in the ear canal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present patent application claims the benefits of priority of U.S. Patent Application No. 62/929,368, entitled “SYSTEM AND METHOD TO PERFORM IN-EAR NOISE DOSIMETRY AND PERSONAL ATTENUATION RATING UNDER AN ELECTRO-ACOUSTIC EARPLUG WHILE EXCLUDING WEARER-INDUCED DISTURBANCES AND SEPARATING EXPOSURE SOURCES” and filed at the United States Patent and Trademark Office on Nov. 1, 2019, the content of which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention generally relates to methods and systems to measure the sound exposure under an electro-acoustic earplug as well as the personal attenuation rating (PAR) provided by the earplug. More specifically, the present invention relates to methods and systems for estimating the in-ear noise exposure from the earplug's receiver and/or the environment while excluding the noise contributions from the wearer (termed “wearer-induced disturbances”). The present invention further relates to methods enabling estimation of the PAR, such as in the context of implementation on embedded computing platforms.

BACKGROUND OF THE INVENTION

(2) Over 12% of the worldwide population is at risk of developing noise induced hearing loss (NIHL). Such proportion represents over 600 million of individuals, of which more than 22 million are North American workers exposed every day to noise exposure doses that put them at risk. Many workers use a variety of hearing protection devices (HPDs) to limit their exposure. Unfortunately, a given HPD can be a poorly suited choice, depending on the initial noise level that it attempts to reduce, if it under-attenuates and lets dangerous noise exposure dose still occur. Furthermore, a given HPD can be inadequate if it over-attenuates, leading to poor situational awareness and communication difficulties that may cause workers to occasionally remove of their HPDs while still subjected to high noise levels, leading to dangerous noise exposure dose. Even if an HPD is theoretically suitable, it can often be improperly fitted and under-perform. Additionally, the background noise level may rise above expectations and make a previously suitable HPD no longer adequate.

(3) Since residual noise level behind the hearing protector is generally unknown, the actual adequateness of the HPD at protecting the worker is generally unknown and, in some cases, the possibility of exposing workers to an excessive noise dose that may lead to permanent hearing damage cannot be ruled out. While in-situ estimation of the PAR helps in determining how the HPD actually performs on a given individual and can possibly assist in training individual to fit their HPD correctly, it is an incomplete solution if the residual noise exposure under the HPD is not considered.

(4) As of now, ensuring that a given individual wearing HPDs does not exceed the recommended maximum noise exposure dose based on an 8-hours shift is difficult. In-ear noise dosimeters featuring an inner ear microphone (IEM) are starting to appear and allow measurement of the residual noise level under the HPD, but they have many limitations. First, they do not allow discrimination of wearer-induced disturbances (speech, chewing noise, etc.) that can contribute significantly to the measured in-ear sound pressure measured inside the occluded ear, notably due to the occlusion effect. While they contribute to in-ear SPL, it is possible that such disturbances may need to be treated differently than other sources due notably to the acoustic stapedius reflex, which reduces hearing sensitivity by up to 15 dB and has been shown to trigger when one speaks. Second, current in-ear dosimeters are not designed to be compatible with advanced hearing protection featuring an in-ear loudspeaker (IELS), which can be used for playback of communication and audio signals and, in combination with an outer ear microphone (OEM), can also be used to offer level dependent attenuation. Third, in-ear dosimeters do not allow segregation of the contribution of the various sound sources that contribute to the in-ear SPL, which can offer

insight to come up with a suitable palliative strategy when the sound exposure limit is exceeded.

(5) There is thus a need for a method or a system to measure the in-ear noise exposure under an advanced HPD while segregating the contributions of wearer-induced disturbances, communication and audio signals and level dependent attenuation.

SUMMARY OF THE INVENTION

(6) The shortcomings of the prior art are generally mitigated by a system and method to measure the sound exposure under an electro-acoustic earplug while excluding wearer-induced disturbances.

(7) In one aspect of the invention, a method to perform in-ear noise dosimetry and personal attenuation rating under an electro-acoustic earplug while excluding wearer-induced disturbances and separating exposure sources using models of electro-acoustical paths to separate the contributions of sound exposure from various sources and provide dosimetry is provided.

(8) The method may further comprise applying different corrections, depending on the type of source, to each of separated contributions to obtain better accuracy by correcting the signals obtained at the IEM so that they better represent the signal that is inside the ear canal, at the inner-end of the ear tip.

(9) In some aspects, the method may comprise using a FIR modeling the transfer function of the earpiece and a source signal provided by an OEM to estimate an IEM signal as a method for WID and playback segregation in the context of dosimetry.

(10) In yet other aspects of the invention, the method may comprise using a FIR modeling of the transfer function between the loudspeaker and the IEM to filter incoming audio signal destined for playback as a method for WID and residual ambient segregation in the context of dosimetry.

(11) The method may further comprise subtracting the estimates of an IEM signal as a method for WID and playback segregation in the context of dosimetry and residual ambient segregation in the context of dosimetry from the IEM to isolate the contribution of WID.

(12) In some aspects of the invention, the method comprises strategies to manage WID contribution to sound exposure in the context of dosimetry, such as including it to the total dose, excluding it, weighting it, or any combination thereof.

(13) In yet other aspects of the invention, the method further comprises temporarily weighting the contribution of all sources when significant WID energy is present to consider the effect of inhibition mechanisms such as the stapedius reflex, triggered by high level WID.

(14) The method may also use the OEM and the IEM estimates to calculate a PAR in real-time that accounts for spectral uncertainty.

(15) In another aspect of the invention, a system to measure noise reduction and evaluate contributions of various sound sources to a sound dose is provided. The system comprise a hearing protection device (HPD) comprising a passive attenuation device adapted to reduce ambient sound present outside an ear of a wearer of the HPD, an inner ear microphone (IEM) adapted to capture a general sound pressure level (SPL) of the sound dose present in the ear canal, outer ear microphone (OEM) adapted to capture the ambient sound, an in-ear loudspeaker (IELS) configured to emit a playback signal in an ear canal of a wearer according to at least one sound source. The system further comprising a processor in signal communication with the IEM and the OEM, the processor being configured to estimate at least one specific SPL corresponding to an identified sound component of the sound dose in the ear canal.

(16) The processor may further be in signal communication with the IELS and the at least one specific SPL is a playback SPL, the processor being configured to compute an estimate value of the playback SPL according to the at least one sound source. The estimate value of the playback SPL may be computed according to at least one IELS-IEM model. The at least one IELS-IEM model may be an adaptive digital filter. The playback signal may be emitted according to a plurality of sound sources.

(17) The at least one specific SPL may be a residual ambient SPL, the processor being configured to compute an estimate value of the residual ambient SPL according to the ambient sound captured

by the OEM. The estimate value of the residual ambient SPL may be computed according to at least one passive attenuation model. The at least one passive attenuation model may be an adaptive digital filter. The processor may be further configured to subtract each of the at least one estimated specific SPL from the general SPL. The remainder of the subtraction may be an estimation of a wearer-induced disturbances (WID) SPL present in the ear canal.

(18) The processor may be further configured to compute a correction for each of the at least one estimated specific SPL. The processor may be further configured to process each of the plurality of sound sources according to a corresponding one of the at least one IELS-IEM model. The plurality of sound sources may be at least one of an audio signal, a communication signal or a level-controlled output of the OEM. The HPD may further comprise an audio mixer adapted to combine the plurality of sound sources.

(19) The HPD may further comprise a digital sound processor configured to receive the playback signal as input. The processor may be further configured to compute an OEM A-weighting according to the ambient sound captured by the OEM and to compute an estimated residual ambient A-weighting according to the estimated residual ambient SPL. The processor may be further configured to compute a personal attenuation rating (PAR) estimate by subtracting the estimated residual ambient A-weighting from the OEM A-weighting.

(20) The system may further comprise a diagnostic module configured to receive the at least one estimated specific SPL. The diagnostic module may be configured to recommend at least one action to be executed by the wearer to reduce the sound dose in the ear canal.

(21) In yet another aspect of the invention, a method to measure noise reduction and evaluate contributions of various sound sources of a sound dose in an ear canal of a wearer is provided. The method comprises capturing the sound dose in the ear canal, the sound dose comprising at least one sound component, estimating a specific canal SPL of the at least one sound component and subtracting the estimated specific canal SPL from the captured sound dose for each of the at least one sound component.

(22) The method may further comprise capturing an ambient sound pressure level (SPL) outside the ear of the wearer and wherein the estimated specific canal SPL is an estimate of the residual ambient SPL present in the ear canal and the estimating is further performed according to the captured sound dose. The estimating may be further performed according to at least one passive attenuation model. The method may further comprise removing the estimated residual ambient SPL from the captured sound dose or may comprise determining an estimation of wearer-induced disturbances (WID) SPL present in the ear canal according to a remainder of the removing.

(23) The method may further comprising emitting an IELS playback signal in the ear canal according to at least one sound source and wherein the estimated specific canal SPL is a playback SPL present in the ear canal and produced by the emitting. The estimating may be further performed by one or more IELS-IEM model. The IELS playback signal may be emitted according to a plurality of sound sources, the estimating comprising estimating one specific canal playback SPL for each of the plurality of sound sources. The estimating may be further performed by one or more IELS-IEM model for each of the plurality of sound sources signal. The method may comprise mixing the plurality of sound sources to generate the IELS playback signal or may comprise removing the estimated IELS playback SPL from the captured sound dose. The method may further comprise estimating of wearer-induced disturbances (WID) SPL present in the ear canal according to the removing. The method may further comprise digitally processing the emitted IELS playback signal.

(24) The method may further comprise emitting an IELS playback signal in the ear canal according to at least one sound source, estimating the playback SPL present in the ear canal according to from the at least one sound sources, removing the estimated residual ambient SPL and the estimated playback SPL from the captured sound dose and estimating wearer-induced disturbances (WID) SPL present in the ear canal according to the removing.

(25) The method may further comprise calculating an OEM A-weighting according to the captured ambient SPL and calculating a canal A-weighting according to the estimated residual ambient SPL. The method may further comprise calculating a personal attenuation rating (PAR) estimate by subtracting the canal A-weighting from the OEM A-weighting. The method may further comprise recommending at least one action to be executed by the wearer to reduce the sound dose in the ear canal according to the estimated specific canal SPL.

(26) In a further aspect of the invention, a system to measure noise reduction and evaluate contributions of residual ambient sound pressure level (SPL) of a sound dose present in an ear canal is provided. The system comprises a hearing protection device (HPD) comprising a passive attenuation device adapted to reduce ambient sound present outside an ear of a wearer of the HPD, an inner ear microphone (IEM) adapted to capture canal sound pressure level (SPL) of the sound dose present in the ear canal, an outer ear microphone (OEM) adapted to capture the ambient sound. The system further comprises a processor in signal communication with the IEM and the OEM, the processor being configured to compute an estimate of the residual ambient SPL of the sound dose in the ear canal.

(27) The estimate of the residual ambient SPL may be computed according to one or more passive attenuation models, the one or more passive attenuation models being determined according to the captured ambient sound. The one or more passive attenuation models may be an adaptive digital filter. The processor may be further configured to perform a subtraction of the estimate of the residual ambient SPL from the canal SPL. An estimation of wearer-induced disturbances (WID) SPL present in the ear canal may be determined according to the subtraction. A calibration of the one or more passive attenuation models may be performed according to the subtraction. The processor may be adapted to execute an adaptive algorithm to calibrate the one or more passive attenuation models according to the subtraction.

(28) In another aspect of the invention, a system to measure noise reduction and evaluate contributions of a playback sound pressure level (SPL) of a sound dose present in an ear canal of a hearing protection device (HPD) wearer is provided. The HPD comprising an inner ear microphone (IEM) adapted to capture a canal sound pressure level (SPL) of the sound dose present in the ear canal, an in-ear loudspeaker (IELS) configured to emit a playback signal in the ear canal, according to at least one sound source. The system further comprising a processor in signal communication with the IEM and the IELS, the processor being configured to estimate a playback SPL according to the canal SPL.

(29) The estimate of the playback SPL may be computed according to at least one IELS-IEM model, the at least one IELS-IEM model being determined according to the at least one sound source. The at least one IELS-IEM model may be an adaptive digital filter. The playback signal may be emitted according to a plurality of sound sources and wherein the processor is further configured to estimate the playback SPL for each of the plurality of sound sources. The processor may be further configured to perform a subtraction of the estimated playback SPL from the canal SPL. An estimation of wearer-induced disturbances (WID) SPL present in the ear canal may be determined according to the subtraction.

(30) A calibration of the at least one IELS-IEM model may be performed according to the subtraction. An adaptive algorithm may be determined for the calibration, the adaptive algorithm being determined according to the subtraction.

(31) Other and further aspects and advantages of the present invention will be obvious upon an understanding of the illustrative embodiments about to be described or will be indicated in the appended claims, and various advantages not referred to herein will occur to one skilled in the art upon employment of the invention in practice.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features and advantages of the invention will become more readily apparent from the following description, reference being made to the accompanying drawings in which:
- (2) FIG. 1 is an illustration of a system to measure noise reduction and evaluate the contributions of various sound sources to noise exposure dose of exposition using electro-acoustic earplugs.
- (3) FIG. 2 a calibration phase/modeling phase of a passive attenuation model according to the principles of the present invention.
- (4) FIG. 3 a calibration phase/modeling phase of an IELS-IEM model according to the principles of the present invention.
- (5) FIG. 4 is an example graph of an amplitude of a response of a passive attenuation model as a function of the frequency according to the principles of the present invention.
- (6) FIG. 5 is example graph of the amplitude of the response of an IELS-IEM model as a function of the frequency according to the principles of the present invention.
- (7) FIG. 6 is an example graph of the amplitude of the correction of a passive attenuation model and of an IELS-IEM model as a function of the frequency according to the principles of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

- (8) A novel method and system to measure noise reduction and evaluate the contributions of various sound sources to noise exposure dose of exposition using electro-acoustic earplugs will be described hereinafter. Although the invention is described in terms of specific illustrative embodiments, it is to be understood that the embodiments described herein are by way of example only and that the scope of the invention is not intended to be limited thereby.
- (9) Now referring to FIG. 1, an embodiment of a system to measure noise reduction and evaluate the contributions of various sound sources to noise exposure dose of exposition using electro-acoustic earplugs **100** is illustrated. The system **100** may be implemented as an advanced HPD in the form of an electro-acoustical earplug. The earplug **1** comprises an OEM **10** and an IEM **12**. In some embodiments, the earplug **1** may further comprise an IELS **14**, generally adapted for playback of audio **4** and communication signals **5**. The IELS **14** may be controlled to lower or increase the volume of sound emitted by the OEM **10**, the audio **4** and communication signals **5**. The earplug **1** may further comprise a processing device, such as a processor or a controller, and a memory unit allowing to store instructions to be executed or any data needed to execute the said instructions.
- (10) The audio **4** and communication **5** signals provide separated contributions to a noise dose received in the ear canal **50**. The earplug **1** is configured to calculate estimates of the various electro-acoustical paths from sound sources to the inner end of the earplug using different models, such as models using adaptive filters.
- (11) In use, the HPD **1** is inserted into the ear canal **50** of a user. The HPD **1** may be configured to have audio playback **4** and/or communication **5** capabilities.
- (12) When worn by the user, a user's sound exposure or sound dose typically comprises three major classes of exposure sources: the residual ambient noise or sounds **52**, sounds **4**, **5** emitted by the IELS **14** and/or wearer-induced disturbances (WID) **56**.
- (13) First, the ambient noise or sound **2** surrounding the ear of the user is attenuated by the HPD **1**. The HPD **1** acts as a passive attenuation **40** resulting in residual ambient noise or sound **52** present in the ear canal **50**.
- (14) Second, the IELS **14** produces an audio playback **4**, a communication signal **5** or any other audio signal. The IELS **14** is typically connected to the OEM **10** which captures the ambient sounds **2**. A level dependent attenuation module **75** is configured to varies the level of the ambient sounds **2** captured by the OEM **10**. In some embodiments, the level dependent attenuated signal **6**,

the audio signals **4** and/or the communication signals **5** may receive additional processing **77** to manipulate the inputted sounds sources through mathematic formulae or algorithms to output respective resulting signals **7**.

(15) The sounds from the different sources, such as the audio playback **4**, a communication signal **5** and/or the level dependent attenuated signal **6**, may be merged, combined or mixed prior to be emitted by the IELS **14**. In some embodiments, the signals **4**, **5** and/or **6** are mix using any mixing device **78**. The resulting mix audio signal **8** is inputted in the IELS which is configured to emit the said signal **8** in the ear canal. The audio signal **8** contributes to the sound exposure or dose of the ear canal **50** in the form of playback sound **54**.

(16) Third, wearer-induced disturbances (WID) **56**, such as talking, chewing or any other wearer inner sounds, contribute to the sound exposure in the form of bone-conducted acoustical signals **56** that are amplified by the occlusion effect present in the ear canal **50**.

(17) The IEM **12** of the HPD **1** is generally positioned in the ear canal **50** to capture the sound exposure or dose within the said ear canal **50**. The output of the IEM **12** is the IEM sound pressure level (SPL) **58**.

(18) The HPD calculates an estimated playback SPL **72** in the ear canal **50** and calculates an estimated residual ambient SPL **62**. In some embodiments, the HPD uses an IELS-IEM model **70** to calculate the estimated playback SPL **72** and a passive attenuation model **60** to calculate the estimated residual ambient SPL **62**. The estimated playback SPL **72** and the estimated residual ambient SPL **62** are subtracted from the IEM SPL **58**. The subtraction process **46** is typically executed using the processor or controller of the HPD **1**. The resulting signal of the subtraction process **46** is an estimated WID SPL **80**.

(19) The resulting estimated SPL **62**, **72** and **80** represents an estimation of the different components of the sound dose within the ear canal **50**.

(20) In some embodiments, the estimated ambient SPL **62** of the contribution of ambient noise **52** is calculated using a passive attenuation model **60**. The passive attenuation models use the SPL captured by the OEM as input. The passive attenuation model **60** typically calculates a transfer function estimate of the noise reduction or attenuation of the HPD **1** using an adaptive digital filter. Such adaptive digital filter models the acoustical path between the OEM **10** and the IEM **12**. Understandably, any known method to evaluate or to compute the attenuation of the HPD **1** may be used within the scope of the present invention.

(21) Referring now to FIG. **2**, in some embodiments, the passive attenuation model **60** may be calibrated to offer a better estimate of the passive attenuation **40** of the HPD **1**. In such embodiments, the adaptive filter is calibrated during an identification or calibration phase. In an exemplary embodiment, the identification phase comprises having ambient sounds being present in the ear canal **50**. Typically, only residual ambient sounds **52** will be present in the ear canal **50** to obtain better calibration results. The residual ambient sounds **52** will be compared to the sounds captured by the OEM **10** to calibrate the passive attenuation model **60**. Understandably, other known calibration methods to calculate the attenuation of the HPD may be used within the scope of the present invention.

(22) This identification or calibration phase may be triggered manually, such as by the user or may be executed periodically at pre-determined time intervals, or executed upon predetermined conditions being detected. Suitable conditions may comprise, but are not limited to, high sound pressure levels being detected by the OEM **10** and low sound pressure levels being detected by the IEM **12**. During calibration, the filter shall converge to a noise reduction model or value associated with the HPD **1**. The calibrated filter is then executed to filter the OEM **10** signal which is dominated by ambient sound. During execution, the calibrated filter outputs an estimate **62** of the portion of sound energy present inside the ear canal **50** due to residual ambient noise **52** using the calculated noise reduction model or value.

(23) Still referring to FIG. **2**, the calibration phase/modeling phase **200** of the passive attenuation

model **60** is illustrated. In such calibration/modeling period **200**, an adaptive algorithm **66** uses the resulting signal $e(n)$ **65** as input to update or recalculate the passive attenuation model **60**. Typically, a known ambient SPL **2** captured by the OEM **10** $x(n)$ is fed to the passive attenuation model **60**. The estimated residual ambient SPL **62** is calculated and is subtracted **46** from the sound dose captured by the OIM **12**. The resulting signal $e(n)$ **65** is used by the adaptive algorithm to calibrate or update the model **60**.

(24) In yet other embodiments, the estimation of the contribution of playback sound **72** to sound exposure relies on a model of the transfer function of the electro-acoustical path from the input of the IELS **14** to the output of the IEM **12** using an IELS-IEM model **70**, such as an adaptive digital filter. The TEL model **70** uses the digitally processed output signals **7** or the audio signals **4**, the communications signal **5** and the level dependent attenuated signal **6** as different inputs. In some embodiments, the IELS-IEM model **70** calculates an estimate of each of the signals **4**, **5**, **6** to produce different playback SPL **72**. Understandably, the present estimation may be configured to use a mixdown signal **8** of the audio signals **4**, **5**, **6** to output a single playback SPL **72**.

(25) Similarly to the passive attenuation model **60**, as illustrated in FIG. **3**, the filter within the model **70** may be calibrated during an identification or calibration phase. Such identification phase generally comprises the step of the IELS **14** producing a test signal **9** in the ear canal **50**. The identification phase may be manually triggered, such as-triggered by the user, may be automatically executed at pre-defined time intervals, or may be automatically executed when suitable conditions are detected. The suitable conditions may comprise, but are not limited to, any signals being currently played back through the IELS **14**, while low sound pressure levels are detected by the OEM **10**. In some embodiments, the estimation model or value may be provided to the filter of the model **70**. As an example, the estimation model or value may be readily available from a communication earpiece adapted to provide echo cancellation. Echo cancellation is generally used to prevent echo in full-duplex communication systems. Echo cancellation generally relies on modeling the path between the input of the IELS **14** to the output of the OEM **10**. The identification phase further comprises accurately calculating the estimation models by letting the filter of the model **70** converging to the estimation models and/or values of the electro-acoustical path between the IELS **14** and the IEM **12**.

(26) Still referring to FIG. **3**, the calibration phase/modeling phase **300** of the IELS-IEM model **70** is illustrated. In such an embodiment, the system **100** further comprises an emitting device **77** for producing a test signal **9**. The test signal **9** is emitted by the loudspeaker **14** in the ear canal and in the TEL model **70** at time of calibration. The IELS-IEM model **70** computes an estimate of the test signal **72** which is subtracted from the sound dose captured by the IEM **12**. The resulting signal **75** ($e(n)$) is processed using an adaptive algorithm or instructions **76** to update or recalculate the IELS-IEM model **70**. As explained above, the resulting signal **75** from the subtraction **46** may be corrected and used in a diagnostic module **49** or may be used during a calibration period.

(27) The calibrated filter is used to filter incoming audio signals, either separately (audio prompts, music, communications, level dependant attenuation), or after combination into an overall audio signal, thereby obtaining an estimate **72** of the portion of sound energy present inside the ear canal **50** due to playback sounds or different estimates **72** for each components of the audio playback signals.

(28) As discussed above, the estimation of the contribution of WIDs **56** is obtained by elimination or subtraction **46**. From the IEM **12** signal, which measures all or most of the sounds present in the ear canal **50**, the predefined portions of sound energy attributable to ambient residual noise **52** and playback sounds **54** may be removed, leaving the portion of sound energy present in the ear canal due to WIDs **56**.

(29) In some embodiments, the HPD or electro-acoustical earpiece **1** may comprise an ear tip having a sound channel (not shown) to which is coupled the IEM **12**. In such embodiments, the IEM **12** is not directly located inside the ear canal **50**, but rather acoustically coupled to the ear

canal **50** through the sound channel. The ear tip or sound channel is adapted be sealingly fitted to the walls of the ear canal **50**. Such seal generally aims at providing a hearing protection. Thus, the IEM **12** signal is not identical to that present inside the ear canal **50**, at the inner end of the ear tip.

(30) In such or in other embodiments, the HPD or electro-acoustical earpiece **1** may be configured to calculate a correction of the estimates SPL outputted by the filters **62**, **72** and/or of the estimated WID SPL **80**. In some embodiments, the correction may differ depending on whether the source of sound exposure is ambient noise **52**, playback sound **54** and/or WIDs **56**. A separate correction can therefore be used for each of these source classes, such as WID contribution correction **22**, IEM playback correction **24** and/or IEM ambient correction **26**. Such corrections **22**, **24** and **26** are applied to the estimated SPL **62**, **72** and **80** to obtain respective corrected signals **64**, **74** and **84**.

(31) Understandably, these corrections **22**, **24** and/or **26** can be pre-calculated using acoustical models, or pre-measured using calibration procedures, and applied separately to each estimated contributive signal to obtain a corrected estimation of the sound pressure level at the inner-end of the ear tip.

(32) From the inner end of the ear tip to the eardrum **51**, the transfer function is independent of the source and a single correction may be used.

(33) The contribution of WIDs **56** to overall noise exposure is still being debated in the literature and, due to mechanisms like the stapedius reflex, may need to be included, reduced by a fixed or variable amount over time, completely excluded, or expressed separately. The present method allows for all these scenarios. In some embodiments, the method comprises temporarily applying one or more weightings in the dose calculation **47** to the contribution of all sources when WIDs **56** are present in significant amount.

(34) In some embodiment, a personal attenuation rating (PAR) **34** may be estimated using the ambient SPL **2** provided by the OEM **10** and the estimate of the residual ambient SPL **62**. In such embodiments, the HPD may be configured to calculate a first A-weighting value **32** of the ambient SPL **2**. The HPD **1** is further configured to calculate a second A-weighting value **30** of the estimated residual ambient SPL **62**. The HPD **1** is further configured to subtract the second A-weighting **30** from the first A-weighting **32** to obtain the PAR estimate **34**. The PAR estimate **34** generally provides another parameter in the analysis of the estimated components **62**, **72** and **80** of the sound dose in the ear canal **50**.

(35) The PAR estimate value **34** generally provides a continuous estimation of the IEM signal that is free of disturbances such as WID **56** or playback sounds **54** that can be used to estimate the PAR, and can capture variations in the PAR **34** that are due to variations in the spectral content of the ambient noise.

(36) In some further embodiments, the estimated SPL **62**, **72**, **80** and/or corrected estimated SPL **64**, **74**, **84** may be inputted to a diagnostic module **49**. The diagnostic module **49** may be configured to calculate suggestion or recommended actions for the user based on the different values of the inputted estimated SPL **62**, **72**, **80** and/or corrected estimated SPL **64**, **74**, **84**. As examples, the diagnostic module **49** may suggest to a user to move to a calmer area as the total sound dose of the ear canal **50** exceeds a predetermined level and the weight of the estimated residual ambient SPL **62** exceed the weights of the other estimated components **72** or **80** or exceeds a predetermined level. In another example, the diagnostic module **49** may suggest to the user to reduce the sound of the audio signal **4** or to suggest reducing the number of calls producing the communication signal **5**. In further other embodiments, the sound levels of the communication signals **5** or audio signals **4** may be automatically reduces when an overall SPL exceeds a predetermined value and the weights of such signal **4**, **5** also exceed one or more predetermined levels.

(37) In some embodiments, the system **100** may further comprise a module to calculate the dose in the ear canal **47** using as input the separated estimated sound dose components **62**, **72**, **80** and/or the corrected values **64**, **74** or **84**. The resulting values may be SPL of each of the components or combined values of each component.

(38) In a further embodiment, the diagnostic module **49** may be fed by standard or best practices in dosimetry. In such embodiment, a comparison module **48** uses as input the separated and/or combined dose calculation **47** and compares the said dose calculation with the standard and best practices. The resulting value may be inference values or boolean values indicating if the said separated or combined doses are within the boundaries of the standard and/or best practices values.

(39) Still referring to FIG. **1**, in yet other embodiments, the system **100** may further comprise a graphical user interface (GUI) **90** adapted to communicate, display and/or read the suggestions or recommendations calculated by the diagnostic module **49**. Understandably, any GUI **90** may be used such as displaying results on a mobile device or a computer, communicating the results through a network or using the loudspeaker **14** to read the suggestion to the wearer.

(40) Another embodiment of the system **100** to measure noise reduction and evaluate the contributions of ambient sound within the ear canal of a wearer is illustrated. The system **100** comprises an OEM **10** and an IEM **12** within the ear canal **50**. The system **100** comprises an earplug or attenuation device **1** having a form of passive attenuation **40**, such as a plug. The system **100** further comprises a DSP or processing device **44**, such as a processor or a controller, and may comprise a memory unit allowing to store instructions to be executed or any data needed to execute the said instructions.

(41) The processing device **44** is configured to calculate and/or compute an estimate value **62** of the residual ambient SPL **52** present in the ear canal **50** based on the ambient SPL **2** captured by the OEM **10**. The estimation is typically computed using a passive attenuation model **60**, such as an adaptive digital filter.

(42) The system **100** further comprises a subtraction module **46** adapted to subtract the estimated ambient SPL **62** from the sound dose captured by the IEM **12** within the ear canal **50**. As explained above, the resulting signal containing the estimated WID SPL **80** from the subtraction **46** may be corrected and used in a diagnostic module **49**.

(43) Yet another embodiment of a system **100** to measure noise reduction and evaluate the contributions of playback sound within the ear canal of a wearer is illustrated. The system **100** comprises an IEM **12** within the ear canal **50** and an IELS **14**, generally adapted for playback of audio **4** and communication signals **5**, as shown in FIG. **1**. The system **100** comprises an earplug or attenuation device, such as a plug. The system **100** further comprises a DSP or processing device **44**, such as a processor or a controller, and may comprise a memory unit allowing to store instructions to be executed or any data needed to execute the said instructions.

(44) As discussed above, the signals provide separated contributions to a noise dose received in the ear canal **50**.

(45) As previously discussed, the processing device **44** is configured to calculate and/or compute an estimate value **72** of the playback SPL **54** present in the ear canal **50** based on the playback source signal, such as audio **4** and communication **5** signals. The estimation is typically computed using an IELS-IEM model **70** for the combined signal **8** (see FIG. **1**) or for each component, such as the audio signal **4** or the communication signal **5**.

(46) Referring now to FIG. **4**, an example graph of the amplitude of the response of the passive attenuation model as a function of the frequency **400** is illustrated. The response for the left **402** and right **404** ears are illustrated.

(47) Referring now to FIG. **5**, an example graph of the amplitude of the response of the IELS-IEM model as a function of the frequency **500** is illustrated. The responses for the left **502** and right **504** ears are illustrated.

(48) Referring now to FIG. **6**, an example graph of the amplitude of the correction of the passive attenuation model **602** and of the IELS-IEM model **604** as a function of the frequency **600** is illustrated.

(49) While illustrative and presently preferred embodiments of the invention have been described in detail hereinabove, it is to be understood that the inventive concepts may be otherwise variously

embodied and employed and that the appended claims are intended to be construed to include such variations except insofar as limited by the prior art.

Claims

1. A system to measure noise reduction and evaluate contributions of various sound sources to a sound dose, the system comprising: a hearing protection device (HPD) comprising: a passive attenuation device adapted to reduce ambient sound present outside an ear of a wearer of the HPD; an inner ear microphone (IEM) adapted to capture a general sound pressure level (SPL) of the sound dose present in the ear canal; at least one of an outer ear microphone (OEM) adapted to capture the ambient sound; and an in-ear loudspeaker (IELS) configured to emit a playback signal in an ear canal of a wearer according to at least one sound source; a processor in signal communication with the IEM and with at least one of the OEM and the IELS, the processor being configured to: estimate at least one specific SPL corresponding to an identified sound component of the sound dose in the ear canal using at least one model of a transfer function between the IEM and at least one of the OEM and the IELS; and subtract each of the at least one estimated specific SPL from the general SPL and wherein a remainder of the subtraction is an estimation of a wearer-induced disturbances (WID) SPL present in the ear canal.
2. The system of claim 1, wherein the at least one specific SPL is a playback SPL, the processor being configured to compute an estimate value of the playback SPL.
3. The system of claim 2, wherein the playback signal combines a plurality of sound sources, the processor being configured to compute an estimate specific SPL for each of the sound sources and to process each of the plurality of sound sources according to the model of the transfer function between the IEM and the IELS.
4. The system of claim 1, wherein the at least one specific SPL is a residual ambient SPL, the processor being configured to compute an estimate value of the residual ambient SPL using the model of the transfer function between the IEM and the OEM and the ambient sound captured by the OEM.
5. The system of claim 1, wherein the processor is further configured to compute a correction for each of the at least one estimated specific SPL.
6. The system of claim 1, wherein the at least one model of the transfer function is an adaptive digital filter.
7. The system of claim 1, wherein the HPD further comprises a digital sound processor configured to receive the playback signal as input.
8. The system of claim 4, the processor being further configured to compute an OEM A-weighting according to the ambient sound captured by the OEM and to compute an estimated residual ambient A-weighting according to the estimated residual ambient SPL.
9. The system of claim 8, wherein the processor is further configured to compute a personal attenuation rating (PAR) estimate by subtracting the estimated residual ambient A-weighting from the OEM A-weighting.
10. The system of claim 1 further comprising a diagnostic module configured to receive the at least one estimated specific SPL.
11. A system to measure noise reduction and evaluate contributions of various sound sources to a sound dose, the system comprising: a hearing protection device (HPD) comprising: a passive attenuation device adapted to reduce ambient sound present outside an ear of a wearer of the HPD; an inner ear microphone (IEM) adapted to capture a general sound pressure level (SPL) of the sound dose present in the ear canal; at least one of an outer ear microphone (OEM) adapted to capture the ambient sound; and an in-ear loudspeaker (IELS) configured to emit a playback signal in an ear canal of a wearer according to at least one sound source; a processor in signal communication with the IEM and with at least one of the OEM and the IELS, the processor being

configured to: estimate at least one specific SPL corresponding to an identified sound component of the sound dose in the ear canal using at least one model of a transfer function between the IEM and at least one of the OEM and the IELS; and compute a correction for each of the at least one estimated specific SPL.

12. The system of claim 6, wherein the plurality of sound sources is at least one of an audio signal, a communication signal or a level-controlled output of the OEM.

13. The system of claim 11, wherein the diagnostic module is configured to recommend at least one action to be executed by the wearer to reduce the sound dose in the ear canal.

14. The system of claim 13, wherein the diagnostic module triggers an audio processing algorithm to automatically reduce specific in-ear SPL caused by one or more of the sound sources.

15. A system to measure noise reduction and evaluate contributions of various sound sources to a sound dose, the system comprising: a hearing protection device (HPD) comprising: a passive attenuation device adapted to reduce ambient sound present outside an ear of a wearer of the HPD; an inner ear microphone (IEM) adapted to capture a general sound pressure level (SPL) of the sound dose present in the ear canal; at least one of an outer ear microphone (OEM) adapted to capture the ambient sound; and an in-ear loudspeaker (IELS) configured to emit a playback signal in an ear canal of a wearer according to at least one sound source; a processor in signal communication with the IEM and with at least one of the OEM and the IELS, the processor being configured to: estimate at least one specific SPL corresponding to an identified sound component of the sound dose in the ear canal using at least one model of a transfer function between the IEM and at least one of the OEM and the IELS; and compute a correction for each of the at least one estimated specific SPL.
